

NeuroNimbus:A Cloud-Powered AI System for Memory Reconstruction

*A Project Report Submitted in the
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COMPUTER SCIENCE AND ENGINEERING

Submitted by

Aare Eesha Sree	23881A05EP
Kandukuri Abhinav	23881A05FR
Katravath Rahul	23881A05FT

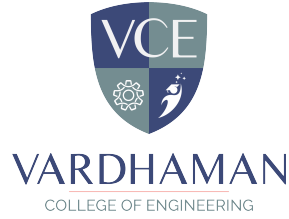
SUPERVISOR

Mr.Syed Ahmeduddin
Assistant Professor



Department of Computer Science and Engineering

April, 2026



Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the project titled **NeuroNimbus:A Cloud-Powered AI System for Memory Reconstruction** is carried out by

Aare Eesha Sree	23881A05EP
Kandukuri Abhinav	23881A05FR
Katravath Rahul	23881A05FT

in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** during the year 2025-26.

Signature of the Supervisor
Mr.Syed Ahmeduddin
Assistant Professor

Signature of the HOD
Dr. Gagandeep Arora
Professor and Head, CSE

Project Viva-Voce held on _____

Examiner

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Aare Eesha Sree

Kandukuri Abhinav

Katravath Rahul

Abstract

NeuroNimbus is a cloud-based intelligent system developed to help people who suffer from memory-related problems such as Alzheimer's or dementia. These individuals often struggle to remember important things like people, events, and daily activities, which makes them depend heavily on caregivers or traditional methods like notebooks and photo albums. However, these manual methods are not well organized and can be difficult to manage over time. To solve this problem, our system provides a smart digital platform that helps in storing and organizing memories in a simple and structured way. It uses two types of interfaces: one for caregivers and one for patients. Caregivers can upload images along with captions and details through a web application, while patients can easily view and access these memories whenever needed. The system uses Natural Language Processing (NLP) to understand the captions provided with each image. It processes the text to extract important keywords that describe the memory. Then, TF-IDF technique is used to identify the most relevant words, and cosine similarity is applied to detect duplicate or similar memories. This helps in avoiding repeated storage and keeps the memory database clean and organized. All memories are grouped based on people and events, which makes it easier for patients to recognize and recall information. The system also includes role-based access control to ensure that only authorized users can access specific data, providing better security and privacy. Since the system is cloud-based, it supports easy storage, fast access, and scalability. The performance of the system was tested and it achieved an overall accuracy of around 90.2%, showing that it works reliably across different functionalities like keyword extraction, categorization, and duplicate detection. In conclusion, NeuroNimbus is a simple, effective, and user-friendly solution that helps improve memory recall for patients while reducing the effort required by caregivers. It combines AI techniques with cloud technology to provide a practical and useful memory assistance system.

Keywords: Digital Memory Reconstruction, Natural Language Processing (NLP), TF-IDF, Cloud Computing

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Abbreviations

Abbreviation	Description
VCE	Vardhaman College of Engineering
CMOS	Complementary Metal Oxide Semiconductor

CHAPTER 1

Introduction

1.1 Introduction

The NeuroNimbus project is a cloud-based intelligent system designed to assist individuals suffering from memory-related disorders such as Alzheimer's disease and dementia. Memory is an essential part of human life, helping people remember past experiences, recognize familiar individuals, and maintain their identity. However, due to cognitive impairments, patients often struggle to recall important information and become dependent on caregivers. Traditional memory aids like notebooks, photo albums, and verbal reminders are mostly manual and can become unorganized and difficult to manage. To overcome these limitations, NeuroNimbus provides a structured and automated digital memory reconstruction system that simplifies memory storage and retrieval .

The significance of this project lies in its ability to improve both patient experience and caregiver efficiency. Many existing systems lack intelligent organization, resulting in duplicate entries, unclear descriptions, and difficulty in accessing relevant information. NeuroNimbus addresses these issues by using Natural Language Processing (NLP) techniques to extract keywords from captions and organize memories into meaningful categories such as people and events. It also uses similarity detection to remove redundant data, ensuring a clean and well-structured memory database. This makes the system highly relevant in healthcare environments where simplicity and usability are important . The main aim of the project is to develop a user-friendly, cloud-supported platform for structured memory management. It includes features such as image upload with captions, keyword extraction, categorization, duplicate detection, and secure storage with role-based access control. The report is organized into sections including introduction, literature review, proposed method, results, and conclusion, providing a complete understanding of the system.

1.2 Background and Motivation

The NeuroNimbus project belongs to the field of healthcare informatics and focuses on cognitive support systems for individuals suffering from memory-related disorders such as Alzheimer’s disease and dementia. Memory is essential for recognizing people, recalling events, and maintaining personal identity. However, due to neurological conditions, patients often face difficulty in remembering important information and become dependent on caregivers. This creates a need for intelligent systems that can assist in managing and reconstructing memories in a structured way .

Traditionally, memory support has relied on manual methods such as journals, photo albums, and audio recordings. Although these methods are useful, they lack organization, automation, and reliability. Over time, they become difficult to manage and may lead to confusion or data loss. With advancements in technologies like Artificial Intelligence, Natural Language Processing (NLP), and cloud computing, digital solutions have emerged. However, many existing systems focus only on specific features like face recognition or voice assistance and do not provide a complete memory management solution .

Previous works such as the Digital Memory Book relied on manual data entry, while systems like Memory Lane used advanced technologies but required high computational resources. Research on NLP techniques like TF-IDF has been effective for text processing but not specifically applied to memory reconstruction.

The motivation behind NeuroNimbus is to overcome these limitations by providing a simple, integrated, and user-friendly system. It combines keyword extraction, categorization, duplicate detection, and cloud storage into a single platform, addressing the gap for an efficient and practical memory assistance solution.

1.3 Objectives of the Project Work

The key aim of this project is to develop a cloud-based intelligent system for structured digital memory reconstruction that assists individuals with cognitive

impairments. The objectives of this work are set out below:

1. To develop a web-based platform that displays photos with captions in a structured layout similar to social media posts.
2. To implement automated person detection and grouping, allowing the system to organize photos of individuals into a dedicated “People” section.
3. To assign and display relationship tags for each identified individual in their profile view for better understanding.
4. To implement smart duplicate memory detection using image similarity algorithms to avoid redundant data.

1.4 Organization of the Report

This report is organized into several chapters, each focusing on a specific aspect of the NeuroNimbus project to provide a clear and structured understanding of the system.

- **Chapter 2:** Literature Survey presents a review of existing memory support systems and related technologies. It discusses previous research works and highlights the limitations that led to the development of this project.
- **Chapter 3:** System Analysis explains the problem domain, system requirements, and the need for an efficient memory reconstruction system. It also describes the challenges faced in existing approaches.
- **Chapter 4:** System Design provides the overall architecture of the proposed system. It explains the workflow, including memory upload, text processing, keyword extraction, categorization, and cloud storage
- **Chapter 5:** System Implementation describes the practical development of the system, including the technologies and methods used such as NLP techniques, TF-IDF, and similarity detection.

- **Chapter 6:** System Testing focuses on validating the functionality of the system. It explains how different modules like keyword extraction, categorization, and duplicate detection are tested for accuracy and performance.
- **Chapter 7:** Results and Discussion presents the experimental results and performance analysis of the system, including accuracy and system behavior under different conditions.
- **Chapter 8:** Conclusions and Future Scope summarizes the overall outcomes of the project and suggests possible improvements and future enhancements to the system.

CHAPTER 2

Literature Survey

2.1 Introduction

The Literature Survey chapter provides an overview of existing research related to digital memory support systems and cognitive assistance technologies. This section is important as it helps in understanding the current developments in the field and identifies the gaps that the NeuroNimbus project aims to address. By reviewing previous studies, we can evaluate their strengths, limitations, and areas where improvement is needed .

The literature review focuses on various approaches used in memory assistance, including manual memory systems, AI-based healthcare applications, Natural Language Processing (NLP) techniques, and cloud-based solutions. The methodology involves analyzing research papers and existing systems to understand how memory-related challenges have been addressed using different technologies. Particular emphasis is given to techniques such as keyword extraction, similarity detection, and structured organization, which are central to the proposed system .

Reviewing prior work is essential because many existing systems focus only on individual features like face recognition or voice assistance and do not provide a complete solution for memory reconstruction. This highlights the need for an integrated system. This chapter discusses relevant studies, compares existing approaches, and identifies research gaps, forming the basis for the NeuroNimbus system.

2.2 Review of Prior Research

This section reviews existing research related to digital memory support systems and AI-based healthcare technologies. Traditional memory support methods such as journals, photo albums, and audio recordings have been

widely used to assist patients with cognitive impairments. However, these methods lack organization, automation, and reliability, making them difficult to manage over time and less effective for long-term use .

Several modern systems have attempted to improve memory assistance using advanced technologies. The Digital Memory Book focused on personalized photo-based memory support but depended heavily on manual input. Memory Lane introduced features like face recognition and voice-based interaction, but it mainly focused on identifying individuals and required high computational resources, limiting its practical usability .

Research in Natural Language Processing (NLP), especially techniques like TF-IDF, has been effective for text processing and keyword extraction. However, these approaches are generally used for general text analysis rather than personalized memory reconstruction. Similarly, cloud computing research has focused on scalability and system design rather than direct application in memory support systems .

Overall, existing works address individual aspects but do not provide a complete solution. This highlights a gap, which NeuroNimbus addresses by integrating multiple functionalities into a single system.

2.3 Identified Research Gaps

This section identifies the limitations and gaps present in existing research related to digital memory support systems. Traditional methods such as journals, photo albums, and audio recordings are mostly manual and lack proper organization and automation. These approaches are time-consuming, prone to errors, and difficult to manage, especially for patients with cognitive impairments and their caregivers .

Modern systems have introduced advanced technologies like face recognition, voice assistance, and AI-based analysis. However, most of these systems focus only on specific functionalities rather than providing a complete solution. For example, some systems emphasize visual recognition, while others focus on text processing or cloud infrastructure. As a result, they fail to offer an integrated framework for structured memory reconstruction. Additionally,

certain solutions require high computational resources, making them unsuitable for real-world healthcare environments with limited infrastructure .

Another major gap is the lack of intelligent organization and redundancy control in existing systems. Many applications do not include automated keyword extraction, similarity detection, or proper categorization, leading to duplicate entries and difficulty in retrieving relevant information.

Therefore, there is a need for a lightweight, integrated, and user-friendly system that combines semantic processing, structured categorization, duplicate detection, and secure cloud storage. NeuroNimbus addresses this gap by providing a unified and practical solution for digital memory reconstruction.

2.4 Summary

In conclusion, the literature survey highlights the current developments and limitations in digital memory support systems and AI-based healthcare technologies. Traditional methods, although useful, lack proper organization and automation, making them less effective for long-term memory management. Modern systems have introduced advanced features such as face recognition, voice assistance, and text processing, but most of them focus only on individual aspects rather than providing a complete solution .

The review also identifies key gaps, including the lack of integrated systems that combine semantic processing, structured categorization, and redundancy control. Additionally, many existing solutions require high computational resources, making them less suitable for practical healthcare environments.

These findings clearly indicate the need for a simple, efficient, and unified system for digital memory reconstruction. The NeuroNimbus project addresses these gaps by integrating keyword extraction, similarity-based duplicate detection, organized memory grouping, and secure cloud storage into a single platform. Thus, this work builds upon existing research and contributes by providing a practical and scalable solution for improving memory assistance systems.

2.5 System Analysis

2.5.1 Existing System

Existing memory support systems mainly rely on traditional methods such as notebooks, photo albums, and verbal reminders. Some modern systems use technologies like face recognition or voice assistants to help patients recall memories. However, these systems are either manual or focus only on specific functionalities. They do not provide a complete solution for organizing, storing, and retrieving memories in a structured way. As a result, memory data often becomes unorganized and difficult to access.

2.5.2 Disadvantages of Existing System

- Lack of proper organization of memory data
- High dependency on manual input
- No intelligent keyword extraction or categorization
- Presence of duplicate or redundant memory entries
- Difficult retrieval of relevant memories
- Some systems require high computational resources
- Not user-friendly for patients with cognitive impairments

2.5.3 Proposed System

The proposed system, NeuroNimbus, is a cloud-based intelligent platform designed for structured digital memory reconstruction. It allows caregivers to upload images with captions, which are processed using Natural Language Processing (NLP). The system extracts keywords, organizes memories into person-based and event-based categories, and detects duplicate entries using similarity measures. It also provides a dual interface for caregivers and patients, ensuring ease of use and better accessibility.

2.5.4 Advantages of Proposed System

- Organized and structured memory storage
- Automated keyword extraction using NLP
- Efficient categorization of memories
- Duplicate detection and removal
- User-friendly interface for patients and caregivers
- Secure cloud storage with role-based access
- Scalable and reliable system performance

2.5.5 System Requirements

Hardware Requirements

- Processor: Intel i3 or above
- RAM: Minimum 4 GB
- Storage: 500 GB or above
- Internet connection for cloud access
- Basic input/output devices (keyboard, mouse, display)

Software Requirements

- Operating System: Windows/Linux
- Frontend: HTML, CSS, JavaScript
- Backend: Python / Node.js
- Database: Cloud-based storage (Firebase / MongoDB)
- Libraries/Tools: NLP libraries for keyword extraction, similarity algorithms

CHAPTER 3

System Design

3.1 System Architecture

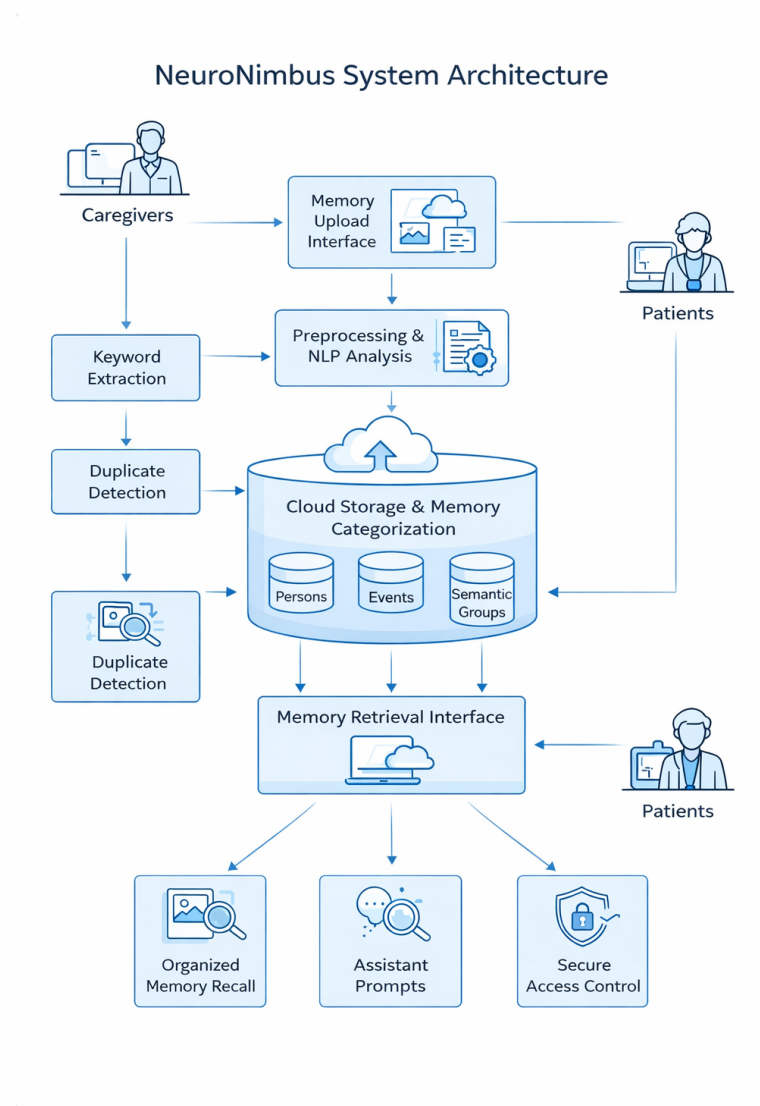


Figure 3.1: System Architecture Diagram.

3.2 Data Flow Diagram

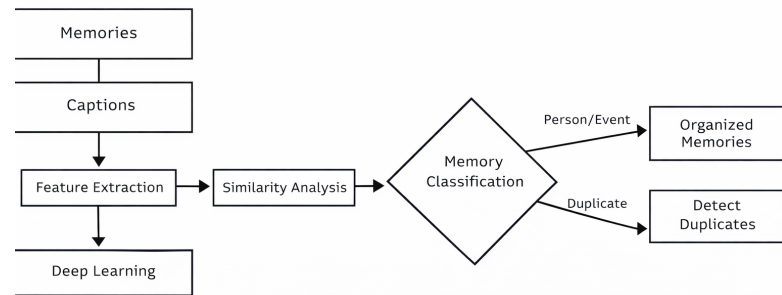


Figure 3.2: Data Flow Diagram

3.3 UML Diagrams

3.3.1 Use Case Diagram

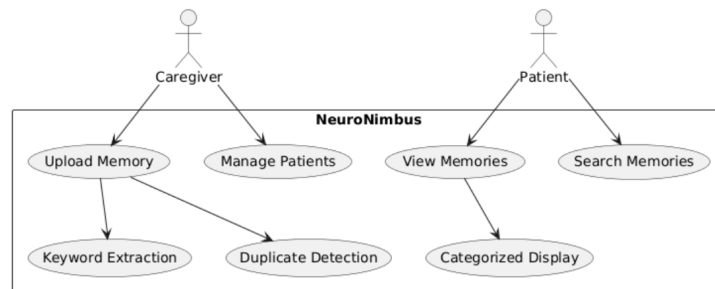


Figure 3.3: Use Case Diagram

3.3.2 Class Diagram

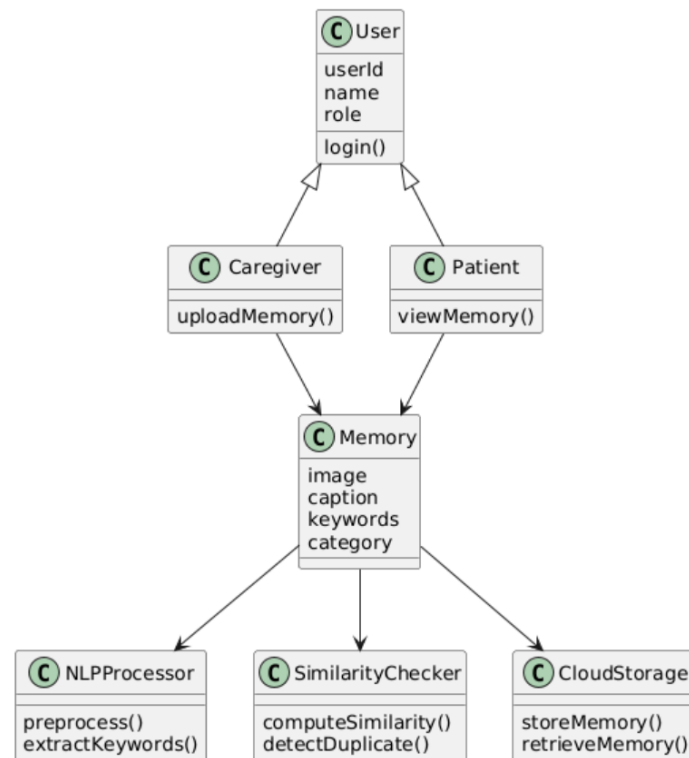


Figure 3.4: Class Diagram

3.3.3 Object Diagram

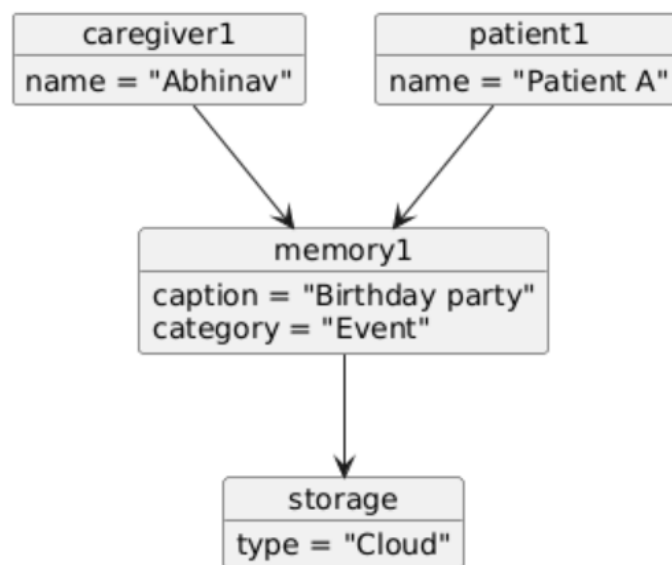


Figure 3.5: Object Diagram

3.3.4 Sequence Diagram

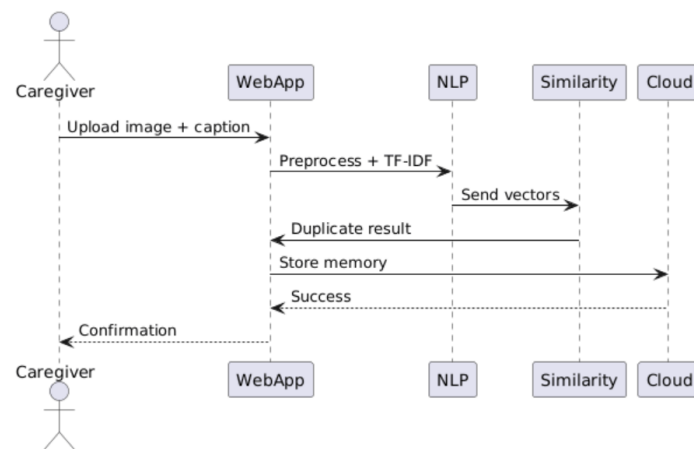


Figure 3.6: Sequence Diagram

3.3.5 Deployment Diagram

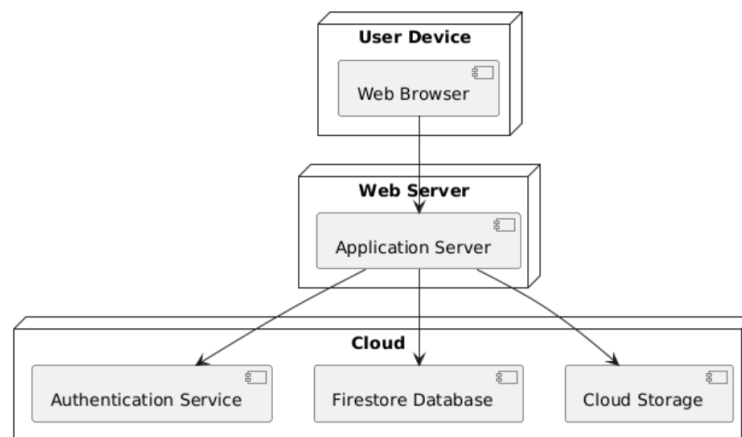


Figure 3.7: Deployment Diagram

3.3.6 Activity Diagram

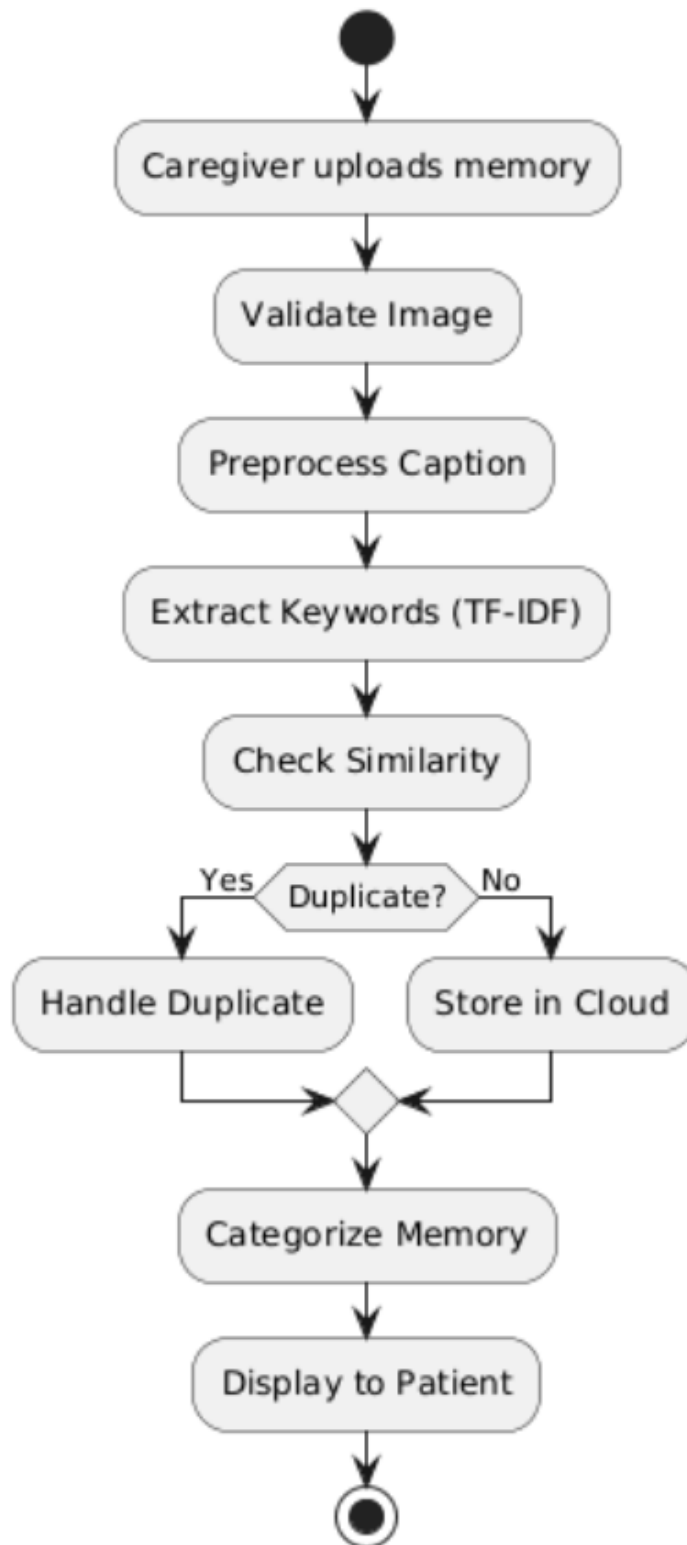


Figure 3.8: Activity Diagram

3.3.7 Component Diagram

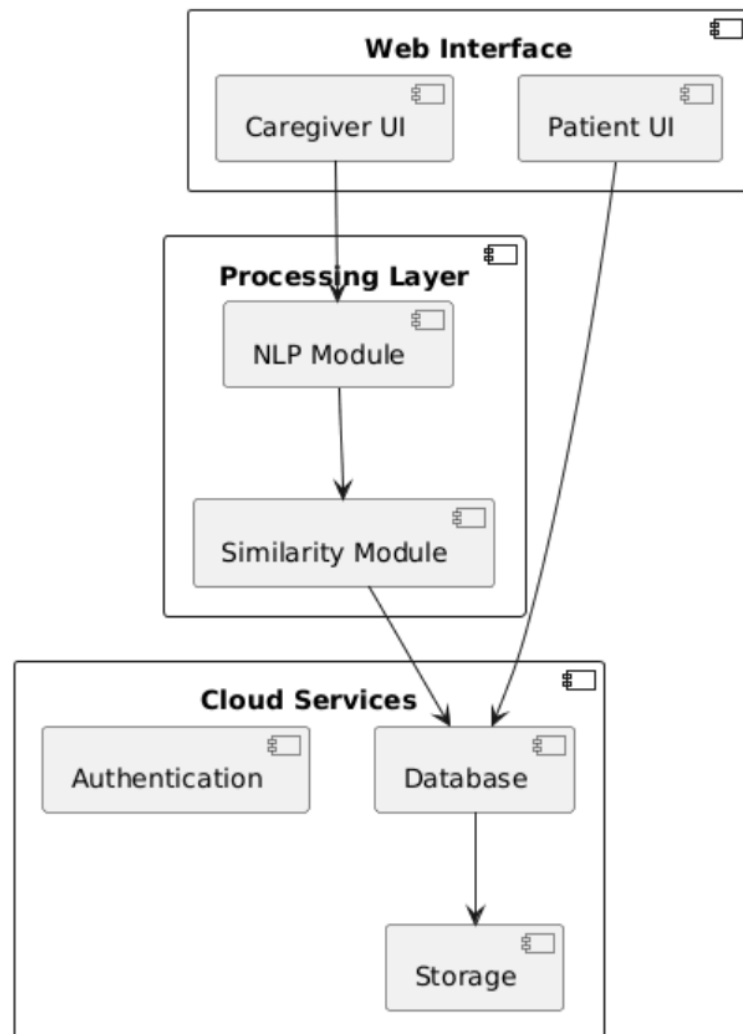


Figure 3.9: Component Diagram

CHAPTER 4

System Implementation

4.1 Modules Description

The NeuroNimbus system is divided into several functional modules to ensure efficient memory processing, storage, and retrieval. Each module performs a specific task and contributes to the overall functioning of the system.

User Management Module This module handles the registration and authentication of users, including caregivers and patients. It ensures secure login and role-based access control, where caregivers can upload and manage data, while patients can only view their memories.

4.1.1 Memory Upload Module

This module allows caregivers to upload images along with captions and additional details such as person names or event information. Ensures that the uploaded data are validated before further processing.

4.1.2 Text Preprocessing Module

In this module, the captions are processed to remove unnecessary words and improve clarity. It includes operations such as lowercasing, tokenization, and stop-word removal to prepare the text for analysis.

4.1.3 Keyword Extraction Module

This module uses Natural Language Processing (NLP) techniques like TF-IDF to extract important keywords from the captions. These keywords help in understanding and organizing memory content.

4.1.4 Memory Analysis Module

This module categorizes memories into person-based and event-based groups. It also detects duplicate entries using similarity measures such as cosine similarity.

4.1.5 Cloud Storage Module

This module stores images, captions, extracted keywords, and category information in a secure cloud environment. It ensures scalability, data security, and efficient data management.

4.1.6 Memory Retrieval Module

This module allows users to search and access stored memories using keywords, categories, or filters. It retrieves relevant data and displays it in an organized and user-friendly format.

4.2 Technology Used

The NeuroNimbus system is developed using a combination of modern web technologies, Natural Language Processing techniques, and cloud-based services to ensure efficient performance and scalability.

4.2.1 Frontend Technologies

The user interface of the system is developed using HTML, CSS, and JavaScript. These technologies help in creating a simple, responsive, and user-friendly interface for both caregivers and patients.

4.2.2 Backend Technologies

The backend of the system is implemented using programming languages such as Python or Node.js. It handles data processing, request handling, and communication between different modules of the system.

4.2.3 Natural Language Processing (NLP)

NLP techniques are used to analyze the text captions associated with images. Methods such as tokenization, stop-word removal, and TF-IDF are used to extract meaningful keywords from the text.

4.2.4 Similarity Detection Techniques

Cosine similarity is used to identify duplicate or similar memory entries. This helps in reducing redundancy and maintaining a clean and organized memory database.

4.2.5 Cloud Technologies

Cloud platforms such as Firebase or MongoDB are used for storing images, captions, keywords, and metadata. Cloud storage ensures scalability, security, and real-time synchronization.

4.2.6 Database Management

The system uses a cloud-based database to manage structured data such as user information, memory records, and categorized data efficiently.

4.2.7 Security Mechanisms

Role-based access control is implemented to ensure that only authorized users can access specific data. This enhances data privacy and system security.

4.3 Algorithms used

4.3.1 Algorithm

Algorithm: Memory Processing and Reconstruction

1. Input memory:

$$M = \{(I_i, C_i)\}_{i=1}^N \quad (4.1)$$

2. Preprocess caption (lowercasing, tokenization, stop-word removal).

3. Compute Term Frequency:

$$TF(t, C_i) = \frac{f(t, C_i)}{\sum_k f(k, C_i)} \quad (4.2)$$

4. Compute Inverse Document Frequency:

$$IDF(t) = \log \left(\frac{N}{df(t)} \right) \quad (4.3)$$

5. Compute TF-IDF:

$$TF-IDF(t, C_i) = TF(t, C_i) \times IDF(t) \quad (4.4)$$

6. Represent caption as vector:

$$V_i = [w_1, w_2, \dots, w_m] \quad (4.5)$$

7. Compute cosine similarity:

$$Sim(V_i, V_j) = \frac{V_i \cdot V_j}{\|V_i\| \|V_j\|} \quad (4.6)$$

8. If

$$Sim(V_i, V_j) > \tau \quad (4.7)$$

mark as duplicate.

9. Categorize memory:

$$F : V_i \rightarrow G \quad (4.8)$$

10. Retrieve memories:

$$R(Q) = \{M_i \mid relevance(V_i, Q) > \delta\} \quad (4.9)$$

Explanation: The algorithm processes memory data using preprocessing,

TF-IDF-based keyword extraction, cosine similarity for duplicate detection, and categorization for structured storage.

4.4 Coding

```
'use client';

import { Button } from '@components/ui/button';
import {
  Card,
  CardContent,
  CardDescription,
  CardHeader,
  CardTitle,
} from '@components/ui/card';
import { Input } from '@components/ui/input';
import { Label } from '@components/ui/label';
import { Logo } from '@components/logo';
import Link from 'next/link';
import { useRouter } from 'next/navigation';
import { useState, type FormEvent } from 'react';
import { signInWithEmailAndPassword, signInWithPopup, GoogleAuthProvider } from 'firebase';
import { useAuth, useFirestore } from '@firebase';
import { useToast } from '@hooks/use-toast';
import { doc, getDoc, setDoc, serverTimestamp } from 'firebase/firestore';

function GoogleIcon() {
  return (
    <svg
      xmlns="http://www.w3.org/2000/svg"
      width="24"
      height="24"
      viewBox="0 0 24 24"
      fill="none"
      stroke="currentColor"
      strokeWidth="2"
    />
  );
}
```

```

        strokeLinecap="round"
        strokeLinejoin="round"
        className="mr-2 h-4 w-4"
    >
    <circle cx="12" cy="12" r="10" />
    <circle cx="12" cy="12" r="4" />
    <line x1="21.17" y1="8" x2="12" y2="8" />
    <line x1="3.95" y1="6.06" x2="8.54" y2="14" />
    <line x1="10.88" y1="21.94" x2="15.46" y2="14" />
</svg>
);
}

export default function LoginPage() {
    const router = useRouter();
    const auth = useAuth();
    const firestore = useFirestore();
    const { toast } = useToast();
    const [email, setEmail] = useState('');
    const [password, setPassword] = useState('');
    const [loading, setLoading] = useState(false);
    const handleLogin = async (e: FormEvent) => {
        e.preventDefault();
        if (!auth) return;
        setLoading(true);
        try {
            await signInWithEmailAndPassword(auth, email, password);
            router.replace('/dashboard');
        } catch (err: any) {
            toast({
                variant: 'destructive',
                title: 'Login failed',
                description: err.message,
            });
        } finally {

```

```

        setLoading(false);
    }
};

const handleGoogleLogin = async () => {
    if (!auth || !firestore) return;
    setLoading(true);
    try {
        const provider = new GoogleAuthProvider();
        const result = await signInWithPopup(auth, provider);
        const user = result.user;
        const userDocRef = doc(firestore, 'users', user.uid);
        const userDoc = await getDoc(userDocRef);
        if (!userDoc.exists()) {
            await setDoc(userDocRef, {
                uid: user.uid,
                displayName: user.displayName,
                email: user.email,
                photoURL: user.photoURL,
                role: 'caregiver',
                createdAt: serverTimestamp(),
            });
        }
        router.replace('/dashboard');
    } catch (error: any) {
        toast({
            variant: 'destructive',
            title: 'Google login failed',
            description: error.message || 'Please try again.',
        });
    } finally {
        setLoading(false);
    }
}

return (

```

```

<div className="flex min-h-screen items-center justify-center p-4">
  <div>
    <Logo className="mb-6 mx-auto" />
    <Card className="w-full max-w-sm">
      <CardHeader>
        <CardTitle>Caregiver Login</CardTitle>
        <CardDescription>
          Login to access the caregiver dashboard
        </CardDescription>
      </CardHeader>
      <CardContent>
        <form onSubmit={handleLogin} className="grid gap-4">
          <div className="grid gap-2">
            <Label>Email</Label>
            <Input
              type="email"
              required
              value={email}
              onChange={(e) => setEmail(e.target.value)}
              disabled={loading}
            />
          </div>
          <div className="grid gap-2">
            <Label>Password</Label>
            <Input
              type="password"
              required
              value={password}
              onChange={(e) => setPassword(e.target.value)}
              disabled={loading}
            />
          </div>
          <Button type="submit" disabled={loading}>
            {loading ? 'Logging in...' : 'Login'}
          </Button>
        </form>
      </CardContent>
    </Card>
  </div>
</div>

```



```

        </Button>
    </form>
    <Button
        variant="outline"
        className="w-full mt-4"
        onClick={handleGoogleLogin}
        disabled={loading}
        type="button"
    >
        <GoogleIcon />
        Login with Google
    </Button>
    <p className="mt-4 text-center text-sm">
        Don't have an account?{' '}
        <Link href="/signup" className="underline">
            Sign up
        </Link>
    </p>
</CardContent>
</Card>
</div>
</div>
);
}

```

4.5 System Testing

4.5.1 Testing Methodology

The NeuroNimbus system is tested to ensure that all modules function correctly and meet the expected requirements. The testing methodology focuses on validating system outputs against expected results. Each module, including keyword extraction, memory categorization, and duplicate detection, is tested individually as well as in integration.

The testing process involves providing sample inputs such as images and captions and verifying whether the system correctly processes them. Functional testing is performed to ensure that each feature works as intended. System-level testing is also carried out to verify smooth interaction between modules, proper cloud synchronization, and secure role-based access control. The results obtained are compared with expected outputs to evaluate the accuracy and reliability of the system.

4.5.2 Testing under Various Stages

The system is tested at different stages to ensure proper functionality:

- **Unit Testing:** Individual modules such as text preprocessing, keyword extraction, and similarity detection are tested separately.
- **Integration Testing:** Interaction between modules is tested, such as the flow of data from memory upload to processing and storage.
- **System Testing:** The complete system is tested to ensure all components work together correctly.
- **User Testing:** The system is tested from the user perspective to check usability for both caregivers and patients.
- **Performance Testing:** The system is evaluated for response time, accuracy, and behavior under multiple user interactions.

4.5.3 Test Cases

S.No	Test Case	Expected Output	Result
1	Upload image with caption	Image and caption stored successfully	Pass
2	Keyword extraction	Relevant keywords extracted correctly	Pass
3	Duplicate detection	Duplicate memory identified	Pass
4	Memory categorization	Memory grouped under correct category	Pass
5	User login (caregiver)	Successful login with upload access	Pass
6	User login (patient)	Access only to view memories	Pass
7	Search using keyword	Relevant memories retrieved	Pass
8	Cloud storage	Data stored and retrieved correctly	Pass

Table 4.1: Test Cases for NeuroNimbus System

Conclusion: The testing results confirm that the NeuroNimbus system performs efficiently with accurate outputs and stable performance across all modules.

CHAPTER 5

Results and Discussion

5.1 Introduction

This chapter presents the results obtained from the implementation and testing of the NeuroNimbus system. The purpose of this section is to evaluate the performance of the system and determine how effectively it meets the objectives of structured digital memory reconstruction. The results are important as they demonstrate the system's ability to perform key functions such as keyword extraction, memory categorization, and duplicate detection with accuracy and reliability .

The outcomes of the system are analyzed based on functional testing and performance evaluation. Each module is assessed to verify whether it produces the expected results under different conditions. The results also highlight how the system improves memory organization and retrieval for users.

This chapter is organized to first present the experimental results, followed by a discussion of system performance, accuracy, and overall effectiveness. The analysis helps in understanding the strengths of the system and its contribution to digital memory assistance.

5.2 Overview of Results

This section presents the key findings obtained from the implementation and testing of the NeuroNimbus system. The system was evaluated using a dataset consisting of memory entries with images, captions, and associated metadata. The results show that the system performs effectively across its main functional modules, including keyword extraction, memory categorization, and duplicate detection.

The keyword extraction module successfully identified important terms from captions, although slight variations were observed when captions contained

incomplete or inconsistent text. The memory categorization module accurately grouped memories into person-based and event-based categories, enabling better organization and accessibility. The duplicate detection module achieved the highest accuracy, efficiently identifying similar or repeated memory entries using cosine similarity.

The experimental results indicate that the system achieved an overall functional accuracy of approximately 90.2%, demonstrating stable and reliable performance. The data also shows that categorization and duplicate detection modules performed more consistently compared to keyword extraction.

Additionally, the system maintained stable cloud synchronization, ensuring secure storage and efficient retrieval of data. Overall, the results confirm that NeuroNimbus provides an effective solution for structured memory management with improved organization and usability.

5.3 Result Screenshots

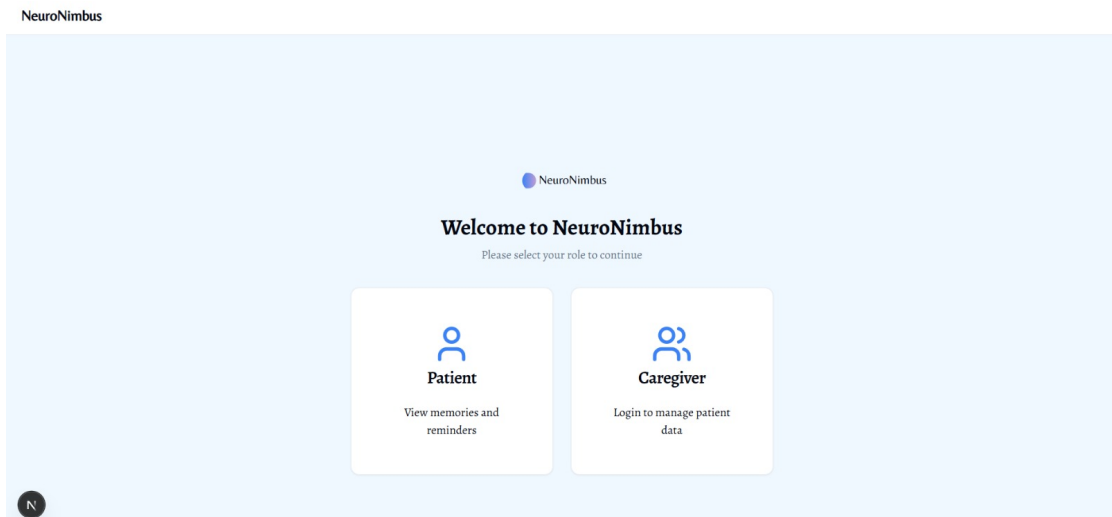


Figure 5.1: User Interface

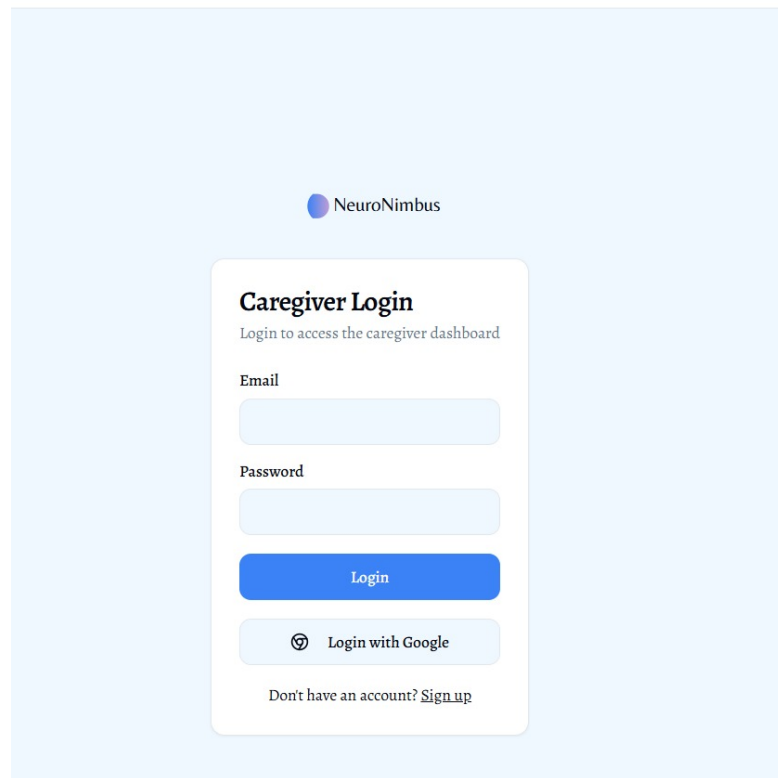


Figure 5.2: Caregiver Login

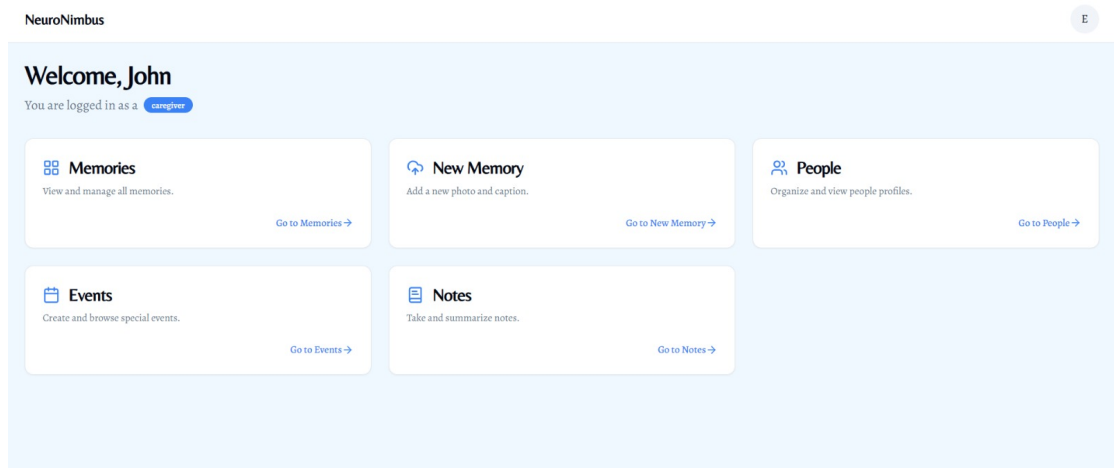


Figure 5.3: Caregiver User Interface

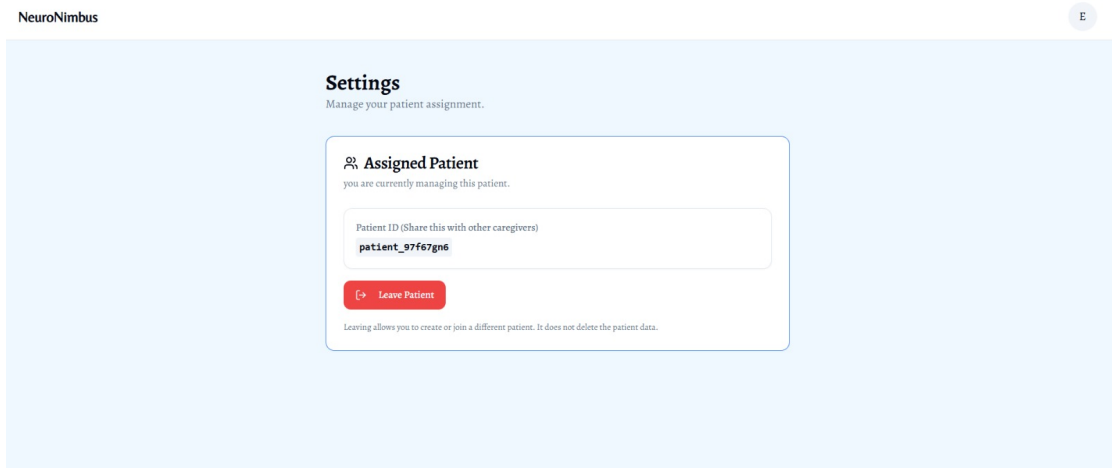


Figure 5.4: Patient ID

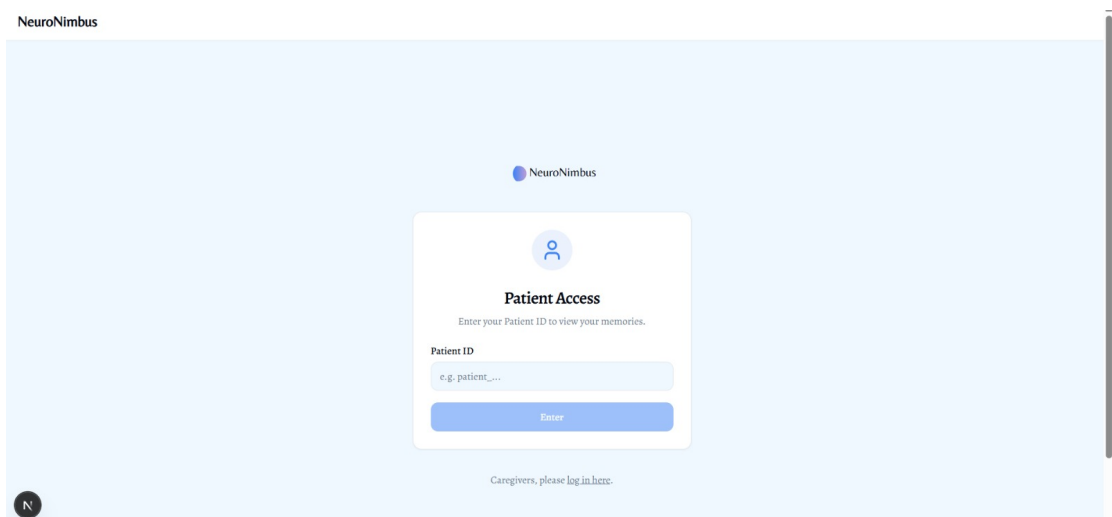


Figure 5.5: Patient Login Interface

5.4 Summary

The NeuroNimbus project is a cloud-based intelligent system designed to assist individuals with memory-related disorders such as Alzheimer’s disease and dementia. These conditions affect a person’s ability to remember important events, recognize familiar people, and manage daily activities. To address this problem, the system provides a structured digital memory reconstruction platform that allows caregivers to upload images along with captions, which are then processed and organized for easy access.

The system uses Natural Language Processing (NLP) techniques to extract meaningful keywords from captions using the TF-IDF method. It also applies

cosine similarity to detect and remove duplicate memory entries, ensuring that the database remains clean and well-organized. Memories are categorized into person-based and event-based groups, making it easier for patients to navigate and recall information.

A dual-interface design is implemented, where caregivers manage data and patients can view memories in a simple and user-friendly format. The system is supported by secure cloud storage with role-based access control, ensuring data safety and scalability.

Experimental results show that the system achieves an overall accuracy of approximately 90.2%, demonstrating reliable performance. Overall, NeuroNimbus provides an efficient, user-friendly, and scalable solution for improving memory recall and reducing caregiver effort.

CHAPTER 6

Conclusions and Future Scope

6.1 Conclusions

In conclusion, the NeuroNimbus project successfully addresses the challenges faced by individuals with memory-related disorders by providing a structured and intelligent digital memory reconstruction system. The project achieves its main objective of developing a user-friendly platform that assists patients in recalling important memories while reducing the effort required by caregivers. By integrating Natural Language Processing techniques such as TF-IDF for keyword extraction and cosine similarity for duplicate detection, the system ensures organized, meaningful, and non-redundant memory storage .

The system also demonstrates effective categorization of memories into person-based and event-based groups, improving accessibility and ease of use. The use of cloud-based storage with role-based access control ensures data security, scalability, and efficient retrieval. The experimental results, with an overall accuracy of approximately 90.2

Overall, NeuroNimbus provides a practical and efficient solution for digital memory assistance. It successfully combines simplicity, functionality, and modern technologies to support cognitive care, making it suitable for real-world healthcare applications.

6.2 Future Scope of Work

The NeuroNimbus system demonstrates effective performance in digital memory reconstruction; however, there are several areas where improvements can be made. One limitation of the current system is its reliance on basic NLP techniques such as TF-IDF, which may not fully capture complex contextual meanings in captions. Future enhancements can include the use of advanced models like transformer-based techniques to improve semantic un-

derstanding. Additionally, improving the robustness of keyword extraction for handling spelling errors and incomplete sentences can further enhance system performance .

Another area of improvement is the addition of speech-to-text functionality, which can help users create memory entries using voice input, making the system more accessible for patients with limited typing ability. The system can also be enhanced by incorporating multilingual support, allowing users to interact with the platform in different languages.

For future scope, the project can be extended by integrating features such as location-based tagging and timeline visualization to provide better context for stored memories. The development of a smart assistant for guiding users during memory retrieval can also improve usability. Furthermore, integration with healthcare systems and wearable devices can expand the application of the system in real-world cognitive care environments.

Overall, these enhancements will make the system more intelligent, accessible, and widely applicable.

REFERENCES

- [1] M. Azam, S. Khalid, S. Almutairi, H. A. Khattak, A. Namoun, A. Ali, and H. S. M. Bilal, "Current trends and advances in extractive text summarization: A comprehensive review," *IEEE Access*, vol. 13, pp. 1–25, 2025.
- [2] R. Che, J. Li, and H. Wang, "Transforming reminiscence therapy: Designing an adaptive VR system based on older adults' emotional connections and narrative patterns," in *Proc. IEEE Int. Symp. Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*, pp. 815–816, 2025.
- [3] J. S., L. S., T. Ragupathi, K. Hariharan, R. Sanjana, and D. G. Arora, "Personalized dementia prediction using machine learning techniques," in *Proc. Int. Conf. Power, Control and Computing Technologies (ICPCT)*, pp. 101–105, 2025.
- [4] D.-G. Lee and S.-B. Lee, "Diagnosis of Alzheimer's disease and frontotemporal dementia from electroencephalography signals," *IEEE Trans. Neural Systems and Rehabilitation Engineering*, vol. 33, pp. 2160–2169, 2025.
- [5] Z.-J. Lin et al., "Dementia assessment using Mandarin speech with an attention-based speech recognition encoder," in *Proc. IEEE ICASSP*, pp. 12461–12465, 2024.
- [6] A. Balamurali et al., "Early detection of dementia: Machine learning for predictive analysis," in *Proc. IDCIoT*, pp. 973–980, 2024.
- [7] C. I. Gonzalez, "Towards a general type-2 fuzzy reasoning approach to modeling modifiable dementia risk factors," in *Proc. ISCFI*, pp. 205–210, 2024.
- [8] S. Routray et al., "Early detection of dementia using a large language model-powered chatbot," in *Proc. PDGC*, pp. 719–724, 2024.
- [9] Y. B. Fadhel et al., "Smart device for monitoring persons with Alzheimer disease," in *Proc. ICAIGE*, pp. 1–6, 2024.
- [10] R. K. Ranjan and D. Rai, "A statistical study of Lewy body and other types of dementia," in *Proc. INSPECT*, pp. 1–6, 2024.
- [11] K. Poonam, R. Guha, and P. P. Chakrabarti, "Artificial intelligence based hierarchical classification of frontotemporal dementia," in *Proc. IEEE EMBC*, pp. 1–4, 2024.
- [12] M. M. Murshid et al., "A robust technique for prediction of dementia using ensemble voting classifier," in *Proc. ICCCNT*, pp. 1–5, 2024.
- [13] K. Y. Chan et al., "Deep neural networks in the cloud: Review, applications, challenges and research directions," *Neurocomputing*, vol. 545, p. 126327, 2023.

- [14] Y. Zhao et al., "Artificial intelligence based multimodal language decoding from brain activity: A review," *Brain Research Bulletin*, vol. 201, p. 110713, 2023.
- [15] M. S. Kabir et al., "Early detection of dementia disease using machine learning approach," in *Proc. ICCCI*, pp. 1–6, 2023.
- [16] P. Reisinho, R. Raposo, and N. Zagalo, "A systematic literature review of virtual reality-based reminiscence therapy," in *Proc. IMET*, pp. 1–8, 2022.
- [17] A. Islam et al., "A predictive analysis for early signs of dementia," in *Proc. ICIT*, pp. 1–5, 2022.
- [18] M. R. Lima et al., "Conversational affective social robots for ageing and dementia support," *IEEE Trans. Cognitive and Developmental Systems*, vol. 14, no. 4, pp. 1378–1397, 2022.
- [19] P.-F. Wu, S.-W. Yen, and K.-Y. Fan, "Investigating the influence through EEG," in *Proc. ECBIOS*, pp. 38–40, 2021.
- [20] A. Bannon, A. Rapeaux, and T. G. Constandinou, "Tiresias: A low cost networked UWB radar system for in-home monitoring of dementia patients," in *Proc. IEEE EMBC*, pp. 7068–7072, 2021.